

MITHIL K GOWDA

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Professional Summary

Cybersecurity graduate specializing in Threat Intelligence, Cyber Threat Analysis, Threat Detection, Adversary Profiling, and AI-driven Cyber Defense. Project Intern at ISRO-LEOS with experience developing secure backend applications using Python, Flask, REST APIs, and MySQL. First author of internationally recognized cybersecurity research focused on intelligent threat detection, adaptive deception, multi-agent cyber defense, and autonomous security architectures. Seeking graduate opportunities in Threat Intelligence, Threat Research, Cyber Defense, and Advanced Threat Analytics.

Education

M.S. Ramaiah University of Applied Sciences, Bengaluru

2022–2026

B.Tech. in Information Science and Engineering

CGPA: **8.80 / 10**

Technical Skills

Threat Intelligence: Cyber Threat Intelligence (CTI), Threat Intelligence Lifecycle, Threat Hunting, Threat Detection, Threat Analysis, Threat Actor Profiling, Indicators of Compromise (IOC), Indicators of Attack (IOA), Tactical Threat Intelligence, Strategic Threat Intelligence

Threat Intelligence Frameworks: MITRE ATT&CK, Cyber Kill Chain, Diamond Model, Intelligence Lifecycle, TTP Mapping

Cybersecurity: MITRE ATT&CK Framework, Cyber Kill Chain, TTP Analysis, Security Monitoring, Incident Response Fundamentals, Vulnerability Assessment, Adaptive Deception, OWASP Top 10

Machine Learning: XGBoost, CNN-LSTM, Deep Q-Network (DQN), Reinforcement Learning, Feature Engineering, Classification Algorithms

Programming: Python, Java, JavaScript, SQL, HTML, CSS, Bash

Security Tools: Wireshark, Nmap, Burp Suite, OWASP ZAP

Networking: TCP/IP, DNS, HTTP/HTTPS, SSH, Network Traffic Analysis

Cryptography: AES-256, SHA-256, Zero-Knowledge Proofs (ZKP)

Databases: MySQL, MongoDB

Operating Systems: Linux, Windows

Professional Experience

Indian Space Research Organisation (ISRO) – Laboratory for Electro Optics Systems (LEOS), Bengaluru
2026

Project Intern – Secure Software Development & Data Analytics

- Designed and developed secure backend applications using Python, Flask, REST APIs, and MySQL supporting secure engineering systems and operational analytics.
- Implemented secure authentication, authorization, database security controls, and API communication supporting enterprise software reliability.
- Developed scalable data processing pipelines supporting structured security analytics, operational intelligence, and data-driven decision support.
- Engineered maintainable backend architectures emphasizing secure software engineering, operational resilience, and data-driven security analysis.

Threat Intelligence Projects & Research

A Multi-Agent AI Cyber Defense Framework Using HoneyBee Method

- First Author and Corresponding Author of an AI-driven cybersecurity framework accepted for presentation at ICIICE 2026 (Dubai, UAE) and accepted as a book chapter in the edited volume *Metaheuristic Optimization for Social Good*.
- Designed an intelligent threat intelligence architecture integrating AI-driven threat detection, adversary behavior analysis, distributed threat intelligence, adaptive deception, and autonomous incident response.
- Developed machine learning models integrating XGBoost, CNN-LSTM, Deep Q-Network (DQN), adaptive honeypots, reinforcement learning, and intelligent adversary behavior analysis.
- Evaluated the framework using CIC-IDS2017 datasets and simulated enterprise attack scenarios, achieving 98.24% detection accuracy, 98.51% precision, 98.10% recall, 0.992 ROC-AUC, and a 1.85% false positive rate.

AI-Governed Carbon Credit Exchange with Digital Carbon Passport

- Abstract accepted at the 4th International Conference on Advancing Sustainable Futures (ICASF 2027), Abu Dhabi University, UAE.
- Designed an intelligent secure platform integrating trust scoring, fraud detection, authentication, transaction validation, and risk assessment for secure digital ecosystems.
- Developed the Secure Trade Authorization and Verification Pipeline (STAVP) utilizing AES-256 encryption, SHA-256 hashing, authentication mechanisms, intelligent transaction validation, and cryptographic security.

C³T-STAVP Cryptographic Framework (Ongoing Research)

- Designing an intelligent cybersecurity framework integrating Character Chaffing & Transposition Technology (C³T), Secure Transaction Authorization & Verification Protocol (STAVP), adaptive trust validation, and resilient cyber defense.
- Researching Zero-Knowledge Proofs (ZKP), post-quantum cryptography, semantic obfuscation, privacy-preserving architectures, and resilient threat intelligence and protection mechanisms
- Developing intelligent cryptographic security mechanisms supporting adaptive cyber defense, distributed trust management, and secure enterprise environments.

Professional Certifications

- IBM IT Fundamentals for Cybersecurity Specialization – IBM
- Ethical Hacker – Cisco Networking Academy
- Systems and Usable Security (Elite) – NPTEL
- Introduction to Cyber Security – IGNOU (SWAYAM)
- Google Cloud Computing – NPTEL
- Introduction to Internet of Things (Silver Medal) – NPTEL
- Business Intelligence and Analytics – NPTEL
- IBM Front-End Developer Specialization – IBM

Achievements

- Selected as Project Intern at the Indian Space Research Organisation (ISRO – LEOS).
- First Author of internationally recognized research focused on AI-driven cybersecurity, intelligent distributed systems, autonomous cyber defense, and secure software architectures.
- Awarded the Reliance Foundation Undergraduate Scholarship (2026).
- Recipient of NPTEL Silver Medal in Introduction to Internet of Things.